1. Cooper & McGillem book

1-10.1 (b), 1-10.2 (a), 1-10.4 (b)

2. An urn contains 5 red, 3 orange, and 2 blue balls. Two balls are randomly selected simultaneously.

(a) What is the sample space?

(b) Define an RV X as the number of orange balls selected. Show the map from the sample space to X.

(c) Find the PMF of X.

(d) Find the CDF of X.

3. Suppose a coin having probability 0.7 of coming up heads is tossed 3 times. Let X denote the number of heads that appear in the 3 tosses. Find the PMF of X.

4. Suppose the CDF of X is given as follows. Find the PMF.

$$F_X(x) = \begin{cases} 0, & x < 1.5 \\ 0.7, & 1.5 \le x < 3 \\ 1, & x \ge 3 \end{cases}$$

Question # 5 is removed from this assignment.

~~5. Let X be a RV with the pdf~~

~~$$f_X(x) = \begin{cases} c(1-x^2), & -1 < x < 1 \\ 0, & otherwise \end{cases}$$~~

~~(a) What is the value of c?~~
~~(b) What is the CDF of X~~
~~(c) What is the probability that 0.5 < X < 2?~~